HEALTHCARE PLATFORM · TECHNICAL OVERVIEW

MediConnect

A production-grade, multi-service healthcare platform connecting patients, doctors, staff, and administrators through a unified, compliant system.

HIPAA Compliant | GDPR Ready | FHIR R4 | Multi-Cloud | Multi-Region

PLATFORM AT A GLANCE

By the Numbers

7
Backend Microservices
5 Node.js + 2 Python

35
FHIR R4 Resource
Types Supported

10
Medical Terminology
Systems

424+
Test Assertions
Across 24 Files

4
User Roles
Patient · Doctor · Admin · Staff

2
AWS Regions
US-East + EU-Central

4
Lambda Functions
Deployed Both Regions

3
Cloud Providers
AWS · GCP · Azure

SYSTEM ARCHITECTURE

7 Microservices

PORT 8081
Patient Service
Node.js / Express
Patient CRUD, IoT vitals, identity verification, GDPR consent, FHIR, HL7 v2.x

PORT 8082
Doctor Service
Node.js / Express
Doctor profiles, EHR, prescriptions with 3 safety gates, Google Calendar sync

PORT 8083
Booking Service
Node.js / Express
Appointments, Stripe billing, prior auth, insurance eligibility, CPT codes

PORT 8084
Communication
Node.js / Express
Real-time chat, Chime video, AI symptom check, risk prediction, summaries

PORT 8085
Admin Service
Python / FastAPI
User management, audit log viewer, analytics, system health

PORT 8086
Staff Service
Node.js / Express
Shift management, tasks, announcements, staff directory

PORT 8005
DICOM Service
Python / FastAPI
DICOM upload, HIPAA Safe Harbor de-identification, PACS, ImagingStudy

REACT SPA
Frontend
React 18 + TypeScript + Vite 5
Tailwind, shadcn/ui, TanStack Query, Socket.io, Stripe, Capacitor (Android)

TECHNOLOGY STACK

Built for Scale

AUTH & IDENTITY

AWS Cognito multi-region pools · JWT · Rekognition ID · MFA

DATABASE

DynamoDB 35+ tables · **Cloud SQL** EHR · **Redis** cache

STORAGE & ENCRYPTION

S3 patient/DICOM · **KMS** PHI encryption · AES-GCM 256-bit

REAL-TIME

Socket.io chat · **Chime SDK** video · **MQTT** IoT vitals

AI & ML

AWS Bedrock Claude 3 · **Vertex AI** fallback · **Azure OpenAI**

ANALYTICS

BigQuery streaming · 4 datasets · GDPR-aware DML DELETE

PAYMENTS

Stripe Elements · PaymentIntents · Webhooks · Refunds

CI/CD & INFRA

GitHub Actions · Docker · **Cloud Run** backup · K8s HPA

MESSAGING & EVENTS

SQS event bus · **SNS** alerts · **SES** 11 email types

SECURITY & COMPLIANCE

Multi-Layer Protection

HIPAA

PHI encryption at rest (KMS)
Audit logs 7-year TTL
Break-glass emergency access
15-min inactivity auto-logout

GDPR

Strict data residency routing
Append-only consent ledger
Right to erasure (15+ tables)
19-source FHIR data export

SOC 2

Idempotency guard (PI1)
13 CloudWatch metric types
Optimistic locking (CC6)
Availability config (A1)

🔐 Encryption

- PHI fields: KMS envelope encryption (phi:kms: prefix)
- OAuth tokens: KMS-encrypted before DynamoDB storage
- Client-side: AES-GCM 256-bit via Web Crypto API
- In transit: HTTPS + HSTS 1-year with preload
- Region-isolated KMS keys (US key / EU key)

🚨 Breach Detection

- 9 security event types → CRITICAL SNS alert
- >50 PHI accesses / 5 min → HIGH alert
- DICOM HIPAA Safe Harbor de-identification
- PII masking in Winston logs (email, SSN, phone)
- Tiered rate limiting (Redis-backed with fallback)

AI & REAL-TIME FEATURES

Intelligence Layer

AI CIRCUIT BREAKER

⚡ **AWS Bedrock**
Claude 3 Haiku · Primary
PRIMARY

↓ failure / timeout

☁️ **GCP Vertex AI**
First fallback
FALLBACK 1

↓ failure / timeout

🔷 **Azure OpenAI**
Second fallback
FALLBACK 2

↓ total failure

🛡️ **Degraded Response**
Safe fallback — never throws
SAFE

AI-POWERED FEATURES

🩺 **Symptom Checker**
Natural language → triage · 5 req/min limit · PII scrubbed

📷 **Clinical Image Analysis**
Medical image interpretation via base64 input

📊 **Risk Prediction**
Patient risk assessment from symptoms and history

📝 **Consultation Summary**
Auto-generates clinical notes from transcripts

+ IoT vitals via MQTT → BigQuery streaming → real-time alerts
+ Video via Amazon Chime SDK with recording pipeline
+ Identity verification via AWS Rekognition

Standards-First Design

🔗 Interoperability Standards

- FHIR R4 — 35 resource types, CapabilityStatement
- SMART on FHIR STU 2.0 — EHR launch context
- HL7 v2.x — 7 message types (ADT, RDE, ORM…)
- CDS Hooks — 4 services (patient-view, order-select…)
- Blue Button 2.0 — CMS Medicare data exchange
- X12 270/271 — Insurance eligibility verification
- C-CDA 2.1 — Clinical document XML generation
- DICOMweb — WADO-RS, STOW-RS, QIDO-RS (PS3.18)
- US Core IG — 18 profiles, 18 validators

🏥 10 Terminology Systems

RxNorm | SNOMED CT | ICD-10-CM | ICD-11 | LOINC | CVX | MVX | NDC | CPT | HCPCS

🗄️ Data Layer

- 35+ DynamoDB tables across US + EU regions
- BigQuery streaming — 4 analytics datasets
- S3 per-region (patient, DICOM, EHR)
- DynamoDB PITR 35-day + cross-region replication

🧪 Testing

- 229 Node.js assertions (custom harness)
- 129 Python pytest (admin + DICOM services)
- 66 Vitest unit + 10 Playwright E2E tests

PRODUCTION-READY HEALTHCARE INFRASTRUCTURE

MediConnect

A unified, compliant platform trusted by patients, clinicians, and administrators — built for the demands of modern healthcare.

🏗️

Resilient Architecture

Multi-cloud, multi-region. Automatic failover. Primary → backup in 5s.

🔒

Security by Design

HIPAA, GDPR, SOC 2. KMS on every PHI field. 7-year audit trails.

⚡

AI-Augmented Care

3-tier AI circuit breaker. Symptom checking, risk prediction, summaries.

🌐

Open Standards

35 FHIR R4 resources, 10 terminology systems, 9 interop standards.

📱

Omnichannel

Web, Android, real-time video, IoT vitals, SMS & email notifications.

🧪

Quality Assured

424+ assertions. E2E, unit, and verification scripts across all services.